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Workshop on Computation: Theory and Practice 2025 (WCTP2025)

Exploring Emoji-based Synthetic Annotations for Filipino-English Sentiment Analysis

By
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SOFTWARE TECHNOLOGY



Outline

- Introduction and Motivation
- Research Opportunity
- Methodology
- Results and Discussion
- Conclusion
- Q&A Segment

Introduction and Motivation

Sentiment Analysis (SA)

- Involves the investigation of the "**positive** or **negative orientation** that a writer expresses toward some object" (Jurafsky and Martin, 2025)
- Representation of sentiment varies
  - **Categorical** (e.g., POS, NEG, NEU)
  - **Dimensional** (e.g., -1.0 to +1.0)
- Sentiment analysis is used in many **downstream** applications
  - Dialogue systems (Bertero et al, 2016)
  - Financial market analysis (Chan et al, 2017)



Image from https://github.com/srini047/text-based-sentiment-analysis

Sentiment Analysis with Philippine Data

- Philippine data is often **multilingual** and **code-switched** – making it harder to work on unlike with only English data
- Although this hasn't hindered work in the field
  - Philippine-based ISPs customer sentiment (Patacsil et al, 2015)
  - Teaching performance (Pacol and Palaoag , 2021)
  - 2022 Philippine Presidential Election (Macrohon et al, 2022)
  - Product and service reviews (Cosme and De Leon, 2024)
- **Problem**: These studies either resort to…
  - Manual annotation → Time-consuming and expensive process resulting in small datasets
  - Translation (typically to English) → Results in a loss of information

Research Opportunity

Leverage Emojis

- Emojis are employed in textual communications with non-verbal components to convey **moods**, **sentiments**, and **emotions** (Saroufim et al, 2018; Tomihira et al, 2018)
- Studies have explored including emojis as a feature in sentiment analysis, particularly for low-resource languages
  - Japanese (Kimura and Katsurai, 2017; Tomihira et al, 2018)
  - French (Saroufim et al, 2018)
  - Chinese (Lou et al, 2020; Liu et al, 2021)
  - Arabic (Hakami et al, 2021)

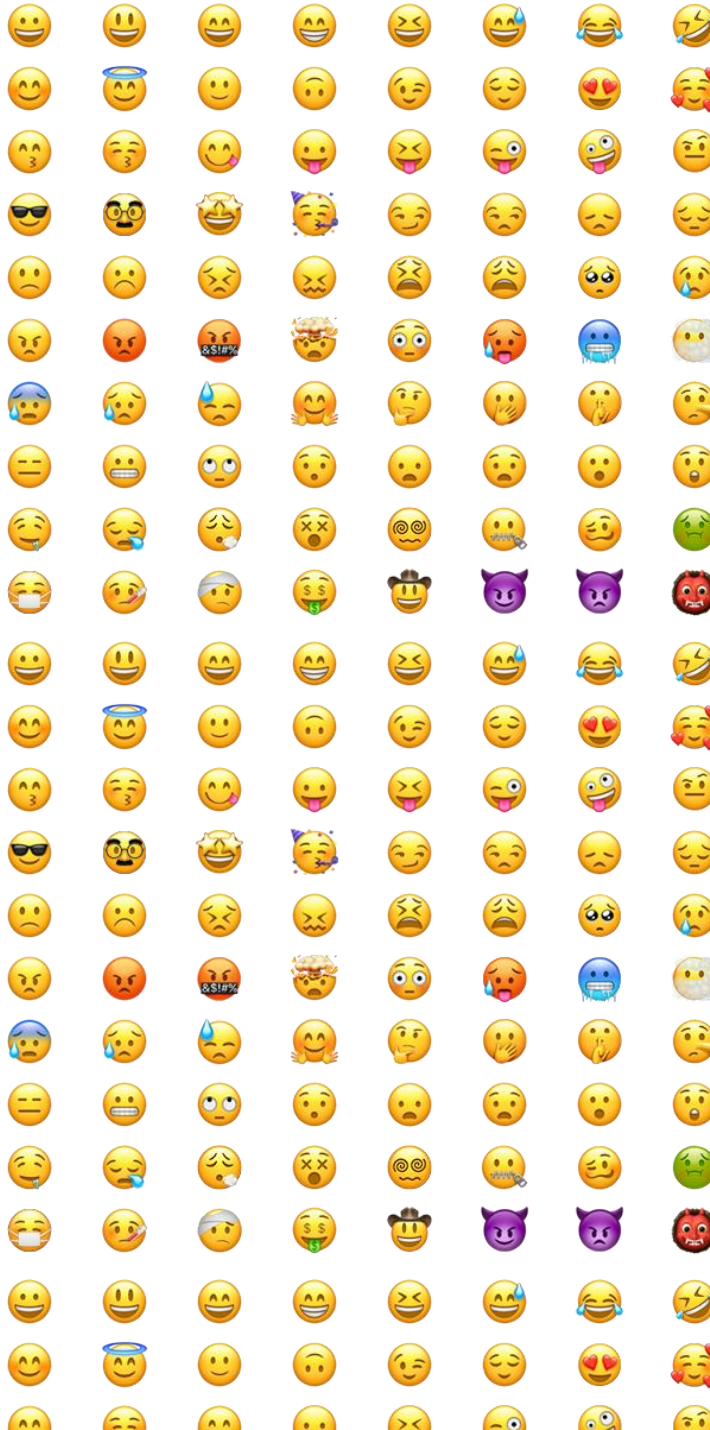
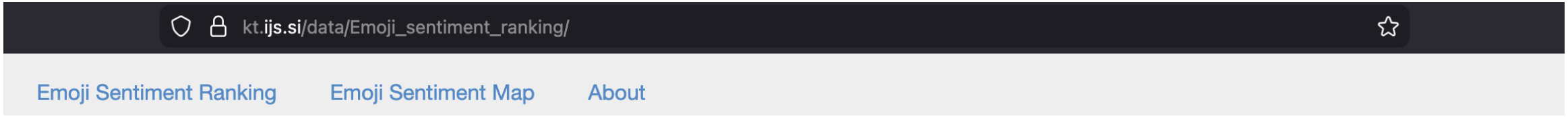


Image from https://seniorplanet.org/ask-the-techspert-emoticons-and-emojis/

Emoji Sentiment Lexicons → Synthetic Annotations?

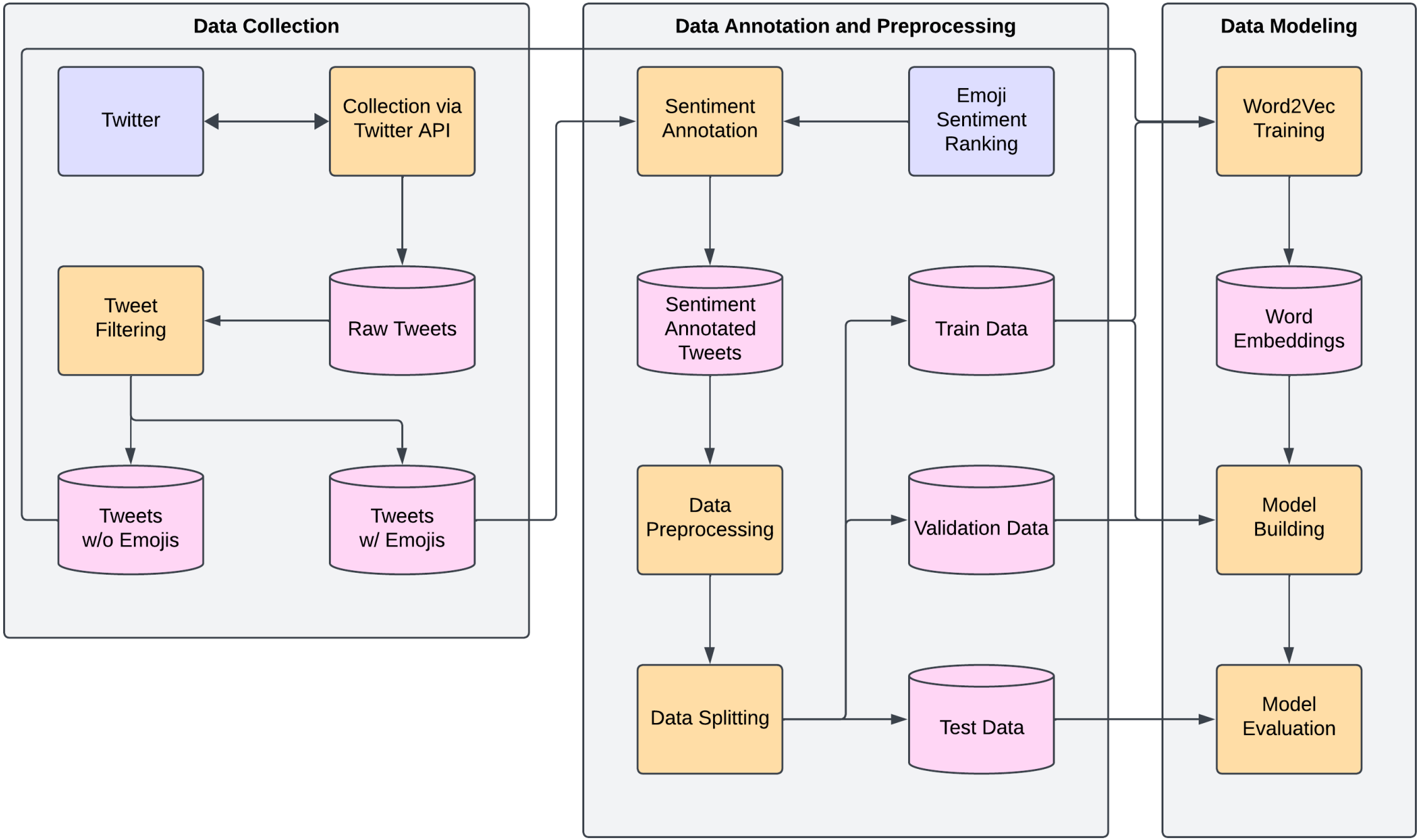
- **Idea:**
  - Leverage existing **emoji sentiment lexicons** to **label** a large volume of Philippine social media text data for **initial training** of a sentiment model
  - Evaluate how well models can **associate sentiment** from **emojis** with the **surrounding words**



Emoji Sentiment Ranking v1.0

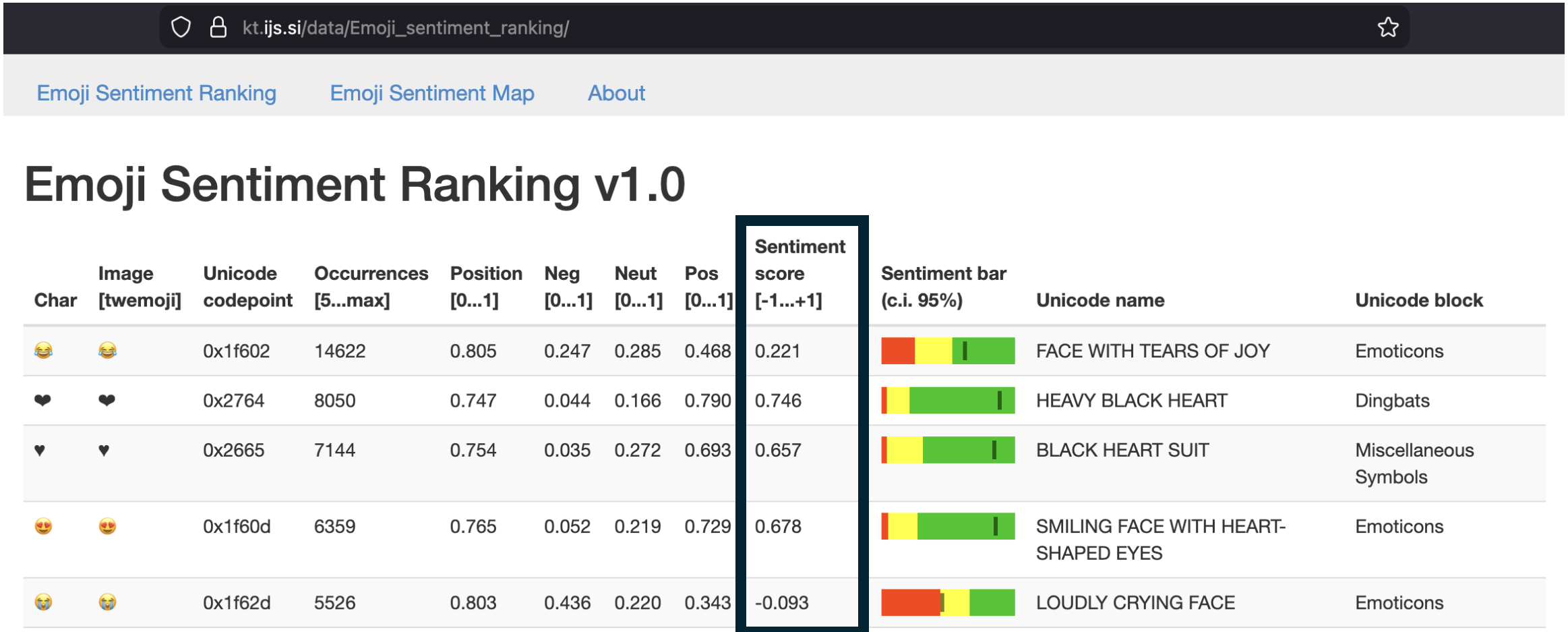
Char	Image [twemoji]	Unicode codepoint	Occurrences [5...max]	Position [0...1]	Neg [0...1]	Neut [0...1]	Pos [0...1]	Sentiment score [-1...+1]	Sentiment bar (c.i. 95%)	Unicode name	Unicode block
😂	😂	0x1f602	14622	0.805	0.247	0.285	0.468	0.221		FACE WITH TEARS OF JOY	Emoticons
❤	❤	0x2764	8050	0.747	0.044	0.166	0.790	0.746		HEAVY BLACK HEART	Dingbats
♥	♥	0x2665	7144	0.754	0.035	0.272	0.693	0.657		BLACK HEART SUIT	Miscellaneous Symbols

Methodology


Data Collection
Twitter
Collection via Twitter API
Tweet Filtering
Raw Tweets
Tweets w/o Emojis
Tweets w/ Emojis
Data Annotation and Preprocessing
Sentiment Annotation
Emoji Sentiment Ranking
Sentiment Annotated Tweets
Train Data
Data Preprocessing
Validation Data
Data Splitting
Test Data
Data Modeling
Word2Vec Training
Word Embeddings
Model Building
Model Evaluation

Our Emoji-based Annotation Scheme

- Utilizes the **Emoji Sentiment Ranking** (ESR) (Kralj-Novak et al, 2015) for sentiment information on Emojis



Emoji Sentiment Ranking v1.0

Char	Image [twemoji]	Unicode codepoint	Occurrences [5...max]	Position [0...1]	Neg [0...1]	Neut [0...1]	Pos [0...1]	Sentiment score [-1...+1]	Sentiment bar (c.i. 95%)	Unicode name	Unicode block
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♥	♥	0x2665	7144	0.754	0.035	0.272	0.693	0.657		BLACK HEART SUIT	Miscellaneous Symbols
😍	😍	0x1f60d	6359	0.765	0.052	0.219	0.729	0.678		SMILING FACE WITH HEART-SHAPED EYES	Emoticons
😭	😭	0x1f62d	5526	0.803	0.436	0.220	0.343	-0.093		LOUDLY CRYING FACE	Emoticons

Our Emoji-based Annotation Scheme

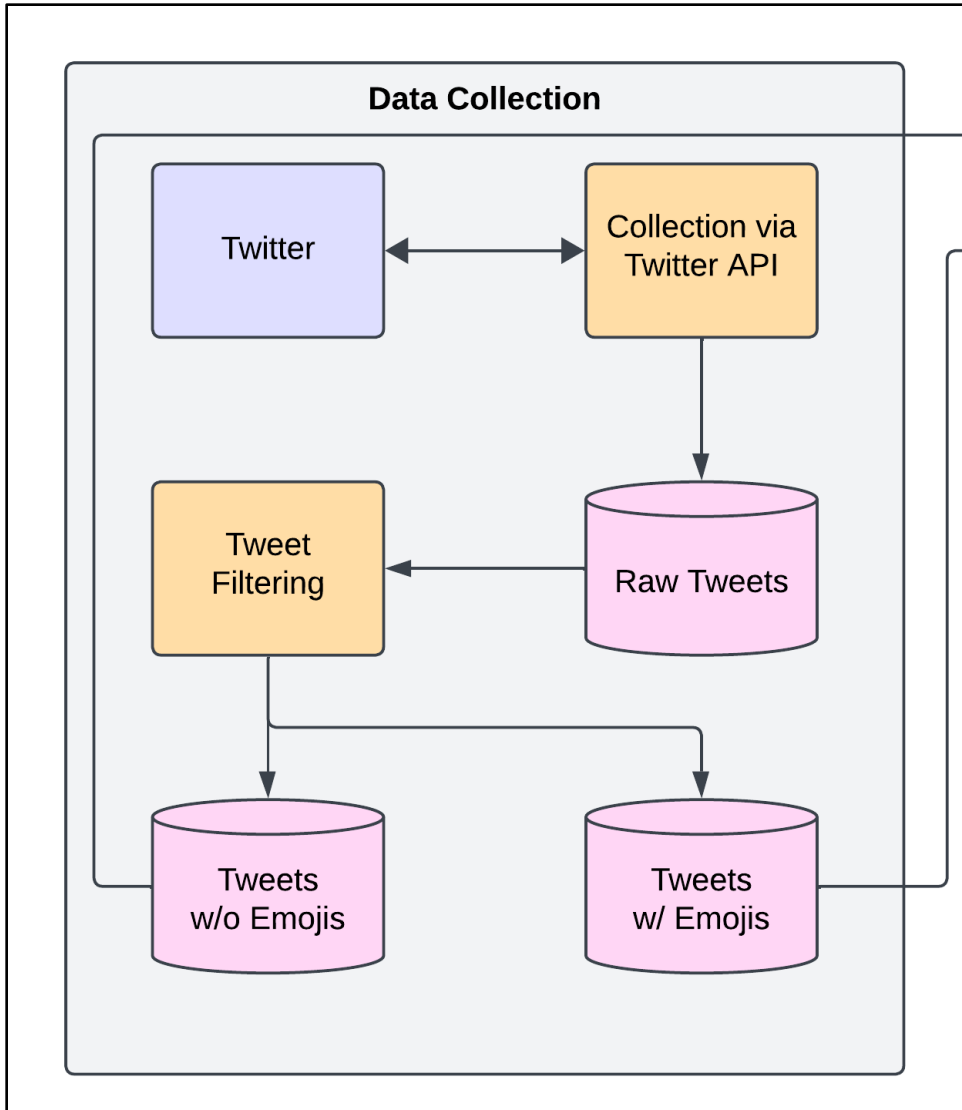
- **Input** → Document that contains at least one emoji present in the ESR
  - Emojis that are not part of the lexicon are ignored
  - Any documents without a recognized emoji are not annotated
- **Processing** → Compute the sentiment score by getting the mean of sentiment scores of all recognized emojis ($\bar{x}$)
- **Output** →
  - If $\bar{x} > 0$, then positive
  - If $\bar{x} < 0$, then negative
  - Else, discard

so funny 😂😭😂 -> [0.221, -0.093, 0.221] -> 0.18 -> POS
galing mo naman hehe 😁💪 -> [0.449, 0.555] -> 0.5 -> POS
NOOO 😡 bawal! 😡 -> [-0.173, -0.173] -> -0.17 -> NEG
Solid movement! 🤖 -> [not recognized] -> n/a -> *discard*

- We limited the possible sentiment labels to **positive** and **negative**

Data Collection

- Utilized **Twitter** API v2 with academic access to collect streamed tweets made in the Philippines between the period of March to December 2020
  - Search filter used a bounding box to define tweets made within the country
- Initially resulted in 10,088,651 tweets
  - Around 2.79 million tweets (27.65%) contained at least one emoji found in the ESR
  - The remaining 72.35% were set aside to train a custom Word2Vec model



Data Collection: Language Distribution

Twitter Language Tag	All Tweets (n=10.09m)	w/ Emojis (n=2.79m)	w/o Emojis (n=7.30m)
Tagalog (TL)	**55.8%**	**47.1%**	**59.1%**
English (EN)	**28.4%**	**34.9%**	**25.9%**
Undefined (UND)	**8.2%**	**11.3%**	**7.0%**
Indonesian (IN)	**3.0%**	**2.6%**	**3.1%**
Spanish (ES)	**0.8%**	**0.7%**	**0.8%**
Others	**3.8%**	**3.4%**	**4.0%**

- Around 80% of the tweets were labeled as either Tagalog or English
- Interestingly, there are more English tweets in the w/ Emoji subset

Sentiment Annotation

- Applying our annotation scheme to the subset of tweets with emojis resulted in…
  - **82.19% positive**
  - **17.78% negative**
  - 0.03% were discarded due to their sentiment scores being exactly zero
- The **high imbalance** of positive to negative instances poses a problem to the learning process; hence, data balancing is applied later on

Data Preprocessing

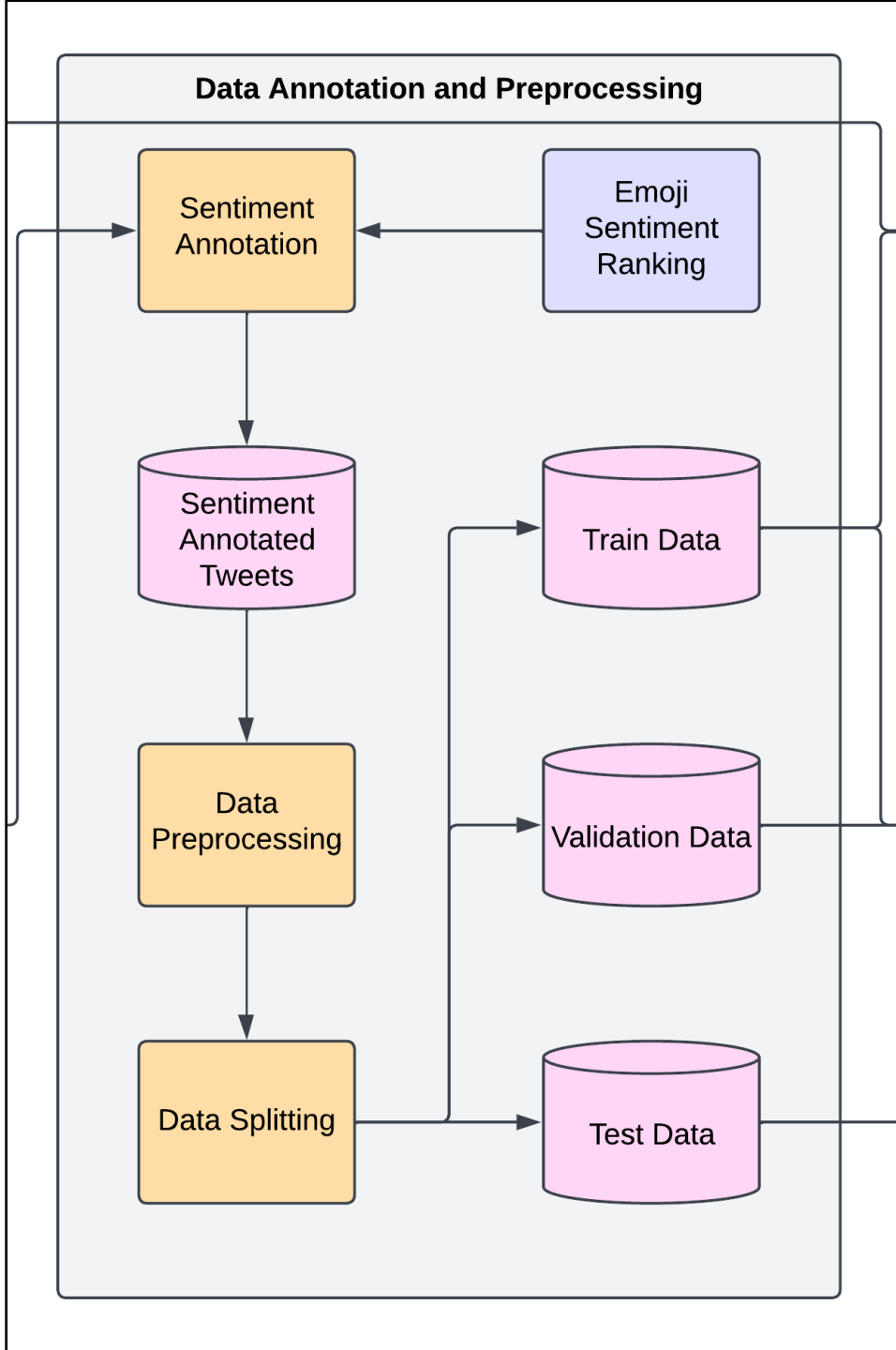
- We built a **custom tokenizer** patterned after existing tokenizers
- Overview of rules:
  - User handles, URLs, time expressions, phone numbers, and generic numbers are each identified and converted to a corresponding placeholder token
    - `@username` → `<USERHANDLE>`
  - Reduce repeating characters greater than or equal to three
    - `Hiiiii` → `Hiii`
  - Lowercase said tokens except those that are entirely in uppercase
    - `Hello` → `hello`
    - `WORLD` → `WORLD`

Data Preprocessing

- After tokenization, the tweets are filtered to retain only those that contain at least one alphanumeric character
- Duplicate tweets were also removed
- The filtering left **2,514,165** tweets for the subset of tweets with emojis

Data Splitting and Balancing

- The dataset was split using random sampling
  - 64% for training (n=1,609,065)
  - 16% for validation (n=402,267)
  - 20% for testing (n=502,833)
- We then applied random undersampling to the positive class due to the 82%-18% class imbalance
  - Removed roughly 77% of the positive data



Data Splitting and Balancing

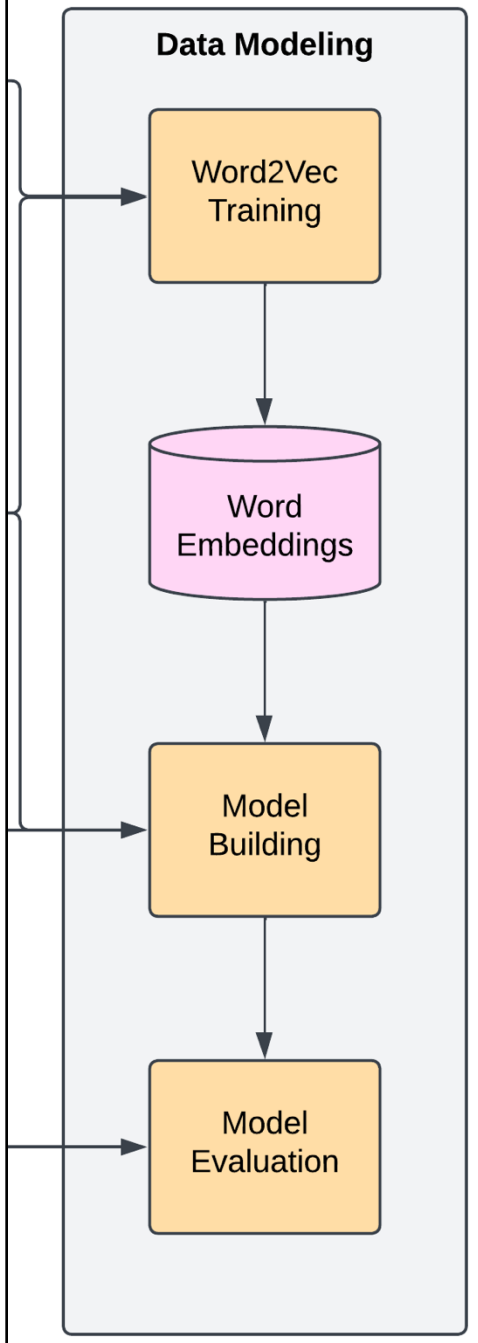
Dataset	Positive	Negative	Total
Train	291,406	291,406	582,812
Validation	72,826	72,826	145,652
Test	90,632	90,632	181,264
Total	**454,864**	**454,864**	**909,728**

Word2Vec Model Training

- While off-the-shelf Word2vec models are available, we opted to **train our own model from scratch** as we have a sizable amount of discarded data
  - Ensure that both English and Filipino words are represented in the embedding space
  - The Word2vec model's domain aligns with the data
- We used a total of **8,922,888 tweets**
  - 7,292,733 rejected tweets (i.e. without emojis)
  - 21,090 tweets that were given a sentiment score of zero
  - 1,609,065 tweets from the entire training dataset before undersampling
- **Descriptive statistics:**
  - 99,016,632 tokens; vocabulary size of 1,583,053
  - Each tweet had an average of 11.1 tokens (±10.2)

Word2Vec Model Training

- We trained a word embedding model using the **skip-gram** implementation in **Gensim** with the following configurations:
  - Vector size: 200
  - Minimum word count: 5
  - Negative sampling: 5
  - NS exponent: 0.75
  - Epochs: 5
  - Window size: 1,2,3
- We trained three models of varying window size to observe the configuration's effect and determined **window size 2** to be the best



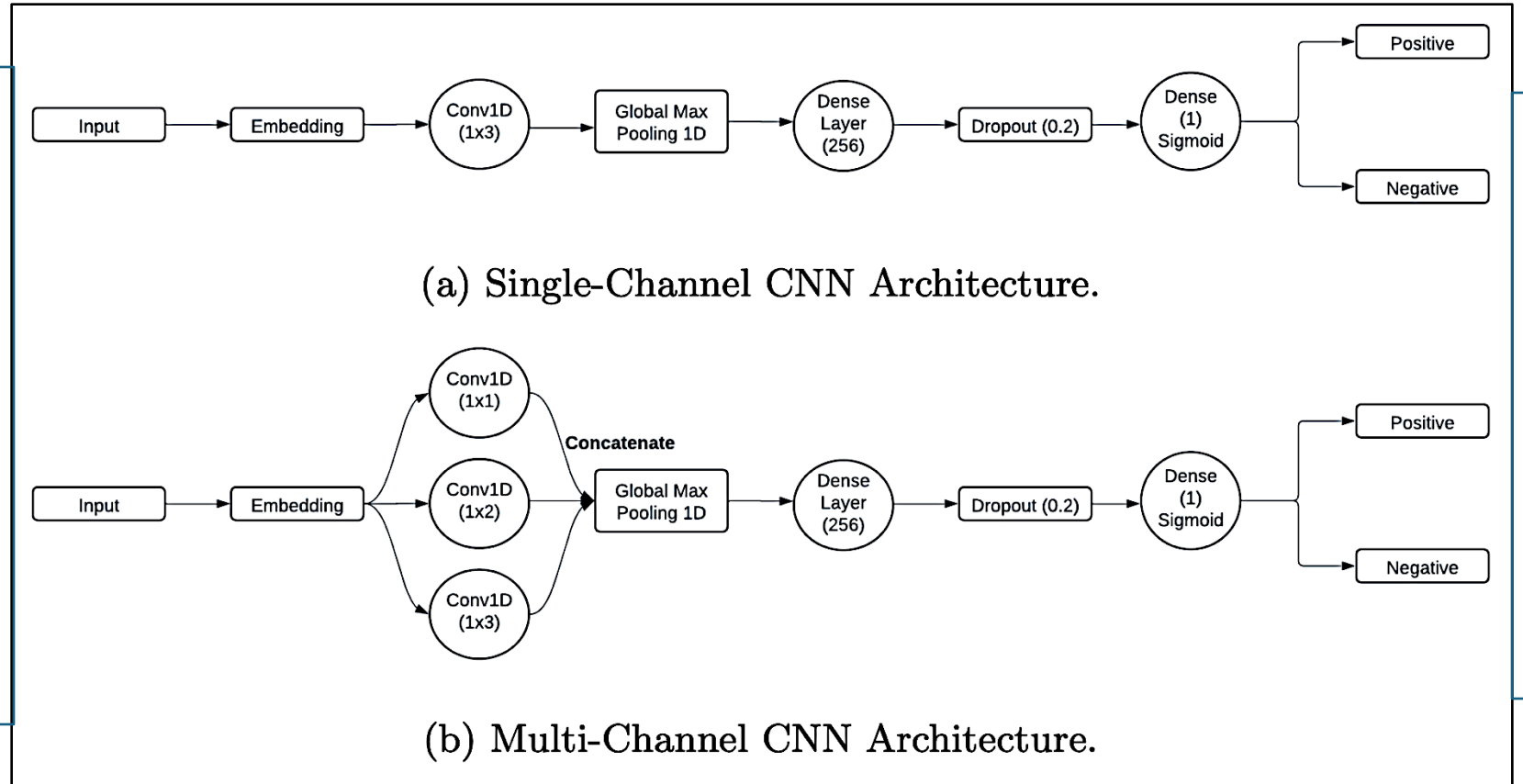
Sentiment Model Training

- Our sentiment classifier is a neural network implemented using Keras designed to classify documents as either positive or negative
- We experimented with two networks structures

Input size: 236
- 236 is max length
- Any tweet shorter was padded

Embedding size: 200

All convolutional layers have 100 filters and utilize a ReLU activation function



(a) Single-Channel CNN Architecture.

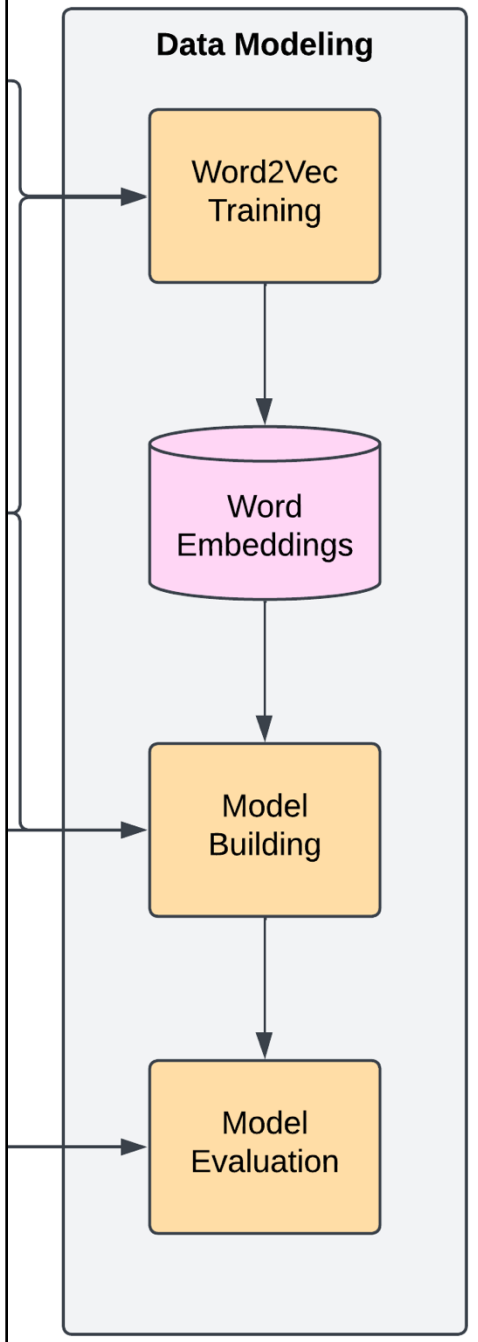
(b) Multi-Channel CNN Architecture.

The network uses the Adam optimizer and binary cross-entropy for the loss function

Based on our initial experiments, we went with the **Multi-Channel CNN Architecture**

Sentiment Model Training

- We performed **hyperparameter tuning** for the **learning rate** and **batch size** of our network
  - The network was trained on 100 epochs with an early stopping mechanism
- The search space for tuning is defined as follows:
  - Batch size: 32, 64, 128, 256
  - Learning rate: 0.00001, 0.0001, 0.001



Test Evaluation

- We report standard metrics such as accuracy, kappa statistic, AUC, and F1-score
- We also look to observe the **effects** of the **inclusion** or **exclusion** of **emojis** from the training and test sets; Defined models as:
  - **M1** → Train with Emoji; Test with Emoji
  - **M2** → Train with Emoji; Test without Emoji
  - **M3** → Train without Emoji; Test with Emoji
  - **M4** → Train without Emoji; Test without Emoji
- Assumptions:
  - Keeping emojis in training set would bias the data
  - An ideal model should associate the sentiment scores from ESR with the words surrounding emojis with or without help from the emoji itself
  - Learn through the noisy, high-volume data

Results and Discussion

Performance Metrics

ID	w/ Emoji?		Train	Test			
	Train	Test	ACC	ACC	Kappa	AUC	F1
M1	Yes	Yes	0.9980	0.9972	0.9943	0.9999	0.9972

- **M1** achieves almost **perfect results**; however, this is **expected** and **misleading** to think this is a good model
  - Sentiment annotations are based on the emoji
  - The model could learn/memorize the label-to-emoji mapping instead of associating sentiment to the surrounding words

Performance Metrics

ID	w/ Emoji?		Train	Test			
	Train	Test	ACC	ACC	Kappa	AUC	F1
M1	Yes	Yes	0.9980	0.9972	**0.9943**	0.9999	0.9972
M2	Yes	No	**0.9152** →	**0.6482**	**0.2964**	0.7618	0.6286

- We confirm the **overfitting** when observing **M2**'s the sharp drop in metrics, particularly with the Cohen Kappa statistic

Performance Metrics

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	Train	Test	ACC	ACC	Kappa	AUC	F1
M1	Yes	Yes	0.9980	0.9972	0.9943	0.9999	0.9972
M2	Yes	No	0.9152	0.6482	0.2964	0.7618	0.6286
M3	No	Yes	0.7741	0.7587	**0.5174**	0.8372	0.7586
M4	No	No	0.7789	0.7666	**0.5331**	0.8464	0.7665

- Models **M3** and **M4** should ideally force the model to associate sentiment from surrounding text rather than emoji tokens
- M4 performs slightly better than M3, indicating that emoji tokens lead to some confusion

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	Train	Test	ACC	ACC	Kappa	AUC	F1
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M3	No	Yes	0.7741	0.7587	**0.5174**	0.8372	0.7586
M4	No	No	0.7789	0.7666	**0.5331**	0.8464	0.7665

- M3 and M4's 0.51-0.53 kappa range shows the models are in **moderate agreement** with emoji-derived sentiment labels
- This suggests that the models are able to **extract sentiment** cues from surrounding text despite **weak supervision**

Limitations & Challenges

- Emoji sentiment may be **contextual** and **culturally dependent**
- **Naive averaging + zero** threshold oversimplifies sentiment
- Binary labels ignore **neutral** or **mixed sentiment**
- Emoji-based labels are **not equivalent** to **human annotation**

Conclusion

Conclusion

- We proposed an **emoji-based sentiment annotation scheme** as a language-independent alternative to manual labeling for Filipino-English text
- Using large-scale Twitter data, we showed that models can **associate text with emoji-derived** despite noisy/weak supervision
  - The most robust model was trained and tested without emojis (kappa of 0.5331)
- Our findings suggest that emoji-based annotation has **potential** for low-resource sentiment analysis, provided emoji leakage is carefully controlled
  - However, more experiments need to be performed before the scheme and learned models can be properly deployed

Future Work

- Our current methodology used a naive approach of averaging scores from the Emoji Sentiment Ranking (ESR) and splitting at a score of 0.0
  - Need to **determine** an **objective** way of **splitting** positive and negative sentiments
  - Also, explore how the **neutral label** can be incorporated into the process
- **Newer** emoji sentiment lexicons, such as the Multidimensional Lexicon of Emojis (MLE) (Godard and Holtzman, 2022), could be explored as alternative sources of sentiment information
- Include additional testing to evaluate the **alignment** of **human judgment** with both the synthetic annotation scheme and the sentiment predictions

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Code and Resource Repository

- To promote **transparency** and **reproducibility**, the resources developed in this study will be made available through the following public repository
  - https://github.com/dlsucelt/PH-Senti-Toolkit
  - Scan QR code to the side →
- Will include:
  - Tweet IDs
  - Pre-trained Word2Vec embeddings and best- performing sentiment classifiers
  - Core preprocessing and training scripts
- **Disclaimer**: Resources are intended exclusively for academic and research purposes



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Question and Answer Segment

- Thank you very much for your attention!
- Do you happen to have any questions? ☺

- For further discussion, feel free to reach out to edward.tighe@dlsu.edu.ph